

COMPUTATIONAL ANALYSIS OF METAL EFFECTS ON ETHYLENE INSERTION IN GROUP 4 METALLOCENE ANALOGS

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INTRODUCTION

ZIEGLER–NATTA POLYMERIZATION

Ziegler–Natta polymerization uses transition-metal catalysts to turn alkenes such as ethylene into polymers. Ethylene first attaches to the metal center and then inserts into the metal–carbon bond, and then makes the chain longer. Metallocene catalysts are useful models for studying this chain-growth step.

BACKGROUND

Metallocene catalysts grow polymer chains by inserting ethylene into a metal–carbon bond. Changing the metal center may affect the bonding, charge distribution, and electronic properties of this step.

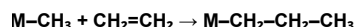
RESEARCH QUESTION

How does changing the metal from Ti to Zr to Hf affect the structure and electronic properties of ethylene insertion?

METHOD

Parameter	Setting
Software	Spartan Student
Method	PM3 semi-empirical
Charge	1
Multiplicity	Singlet
Metals	Ti, Zr, Hf
Stages	Reactant, transition state, product

REACTION SCHEME



TRANSITION-STATE VERIFICATION

Metal	Imaginary frequency
Ti	-319 cm ⁻¹
Zr	-343 cm ⁻¹
Hf	-311 cm ⁻¹

KEY TERMS AND ELEMENT GUIDE

Elements	Metallocene	Transition state	Electronic terms	Catalyst	Imaginary frequency
Ti: Titanium	Metal complex between two ring-shaped carbon ligands	Short-lived structure between reactant and product	HOMO–LUMO gap: energy difference between orbitals	substance that speeds a reaction without being consumed	vibrational result used to support a transition-state assignment
Zr: Zirconium			Dipole moment: charge separation		
Hf: Hafnium					

ANALYSIS

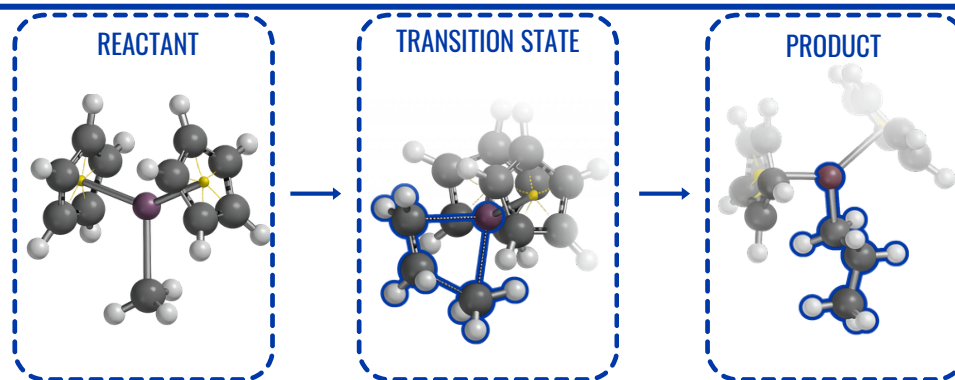
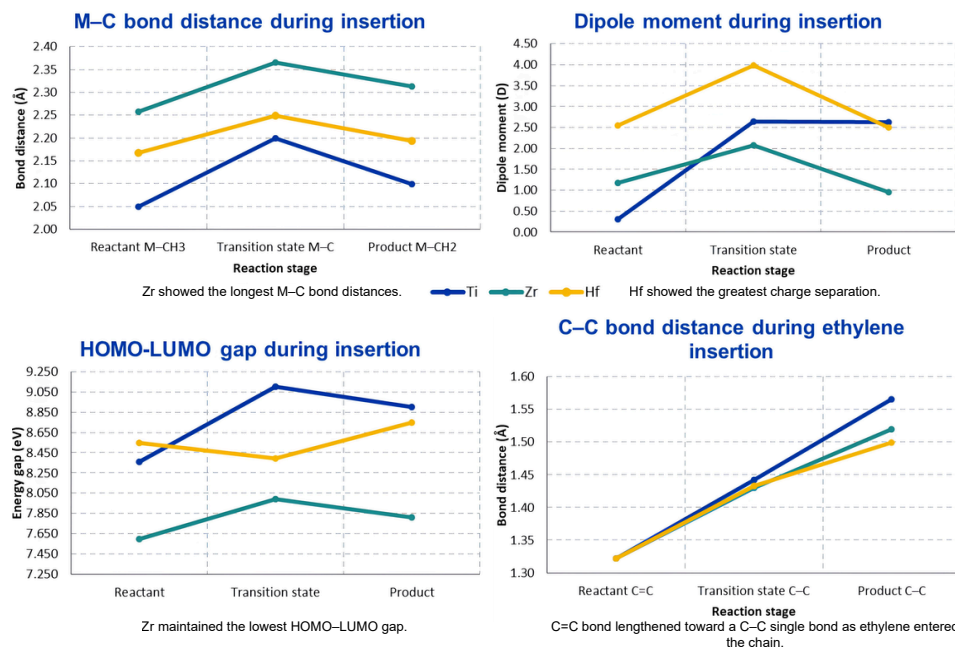


Figure 1. Modeled ethylene-insertion pathway showing the reactant, transition state, and inserted chain product.



TAKEAWAY

MAIN FINDINGS

- Zr**
- Longest M–C bond distances
 - Lowest HOMO–LUMO gap
- Hf**
- Highest dipole moment
 - Greatest charge separation at the transition state
- All three metals**
- Followed the same general ethylene-insertion pathway
 - Formed one validated transition state with one imaginary frequency

LIMITATIONS

- PM3 gives approximate trends rather than highly accurate values.
- Solvent, counterions, catalyst supports, and long polymer chains were not included.
- The calculated energies were not used as activation energies.

CONCLUSION

Changing the Group 4 transition metal changed both the structure and electronic properties of ethylene insertion. Zr showed the longest M–C bonds and the lowest HOMO–LUMO gaps, while Hf showed the strongest polarity.

FUTURE WORK

- Recalculate selected structures using xTB or ORCA.
- Study longer chain-growth steps.
- Connect molecular changes to polymer structure and material properties.

WHY THIS MATTERS

Ethylene insertion is the key step that grows polyethylene chains. Changing the metal center can affect bonding, charge separation, and electronic behavior during chain growth. Understanding these differences can help guide the design of improved polymerization catalysts.

ACKNOWLEDGMENTS

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